Modelling Transient Redistribution of Charge in a Conductor

***Abstract*—complete when done**

I. Introduction

The aim of this project is to arrive at a numerical model for charge redistribution in a conductor at very small time scales. We've been trying to build the model from the foundations up to ensure we understand all the derivation steps and assumptions made, so what we have so far might be a bit too general to be practical, but we've judged that it's more reasonable to simplify our model later rather than earlier.

II. Model

In this section, we introduce all the quantities that constitute the state of the system we are trying to model while also fully defining the time evolution of the system. The system can be described roughly as the sum of three components: the electromagnetic field in (and around) the conductor, the ionic background(bound charges), and the conduction electrons(free charges). We seek to treat the conduction electrons in the conductor as free and damped. We completely disregard the bound charge response as it is negligible in comparison to the free charge response. Additionally, since we treat the conduction electrons as free, there is a static positive ionic background.

A. Electromagnetic field

We simply use Maxwell's equations in vacuum:

$$\begin{aligned}\nabla \cdot \mathbf{E} &= \frac{\rho}{\varepsilon_0}\\ \nabla \cdot \mathbf{B} &= 0\\ \nabla \times \mathbf{E} &= -\partial_t(\mathbf{B})\\ \nabla \times \mathbf{B} &= \mu_0(\mathbf{J} + \varepsilon_0 \partial_t(\mathbf{E}))\end{aligned} \tag{1}$$

B. Charges

In this subsection, we determine the equations that govern the time evolution of the charge distribution in response to the electromagnetic field. At first we treat the free and bound charges separately, taking $\rho = \rho_\mathrm{f} + \rho_\mathrm{b}$.

We start by taking the equation for the Lorentz force

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \tag{2}$$

and applying it to the free charge density distribution:

$$\begin{aligned}\frac{\delta \mathbf{F}}{\delta V} = \frac{\delta q}{\delta V}(\mathbf{E} + \mathbf{v} \times \mathbf{B}) = \rho_\mathrm{f}\mathbf{E} + \mathbf{J} \times \mathbf{B} = \frac{\partial_t(\delta \mathbf{p})}{\delta V} =\\ = \frac{\partial_t(\mathbf{v}\delta m)}{\delta V} = \frac{\partial_t(\mathbf{v}\frac{\delta q}{s})}{\delta V} = \frac{\partial_t(\frac{\mathbf{v}\delta q}{\delta V})}{s} = \frac{\partial_t(\mathbf{J})}{s},\end{aligned} \tag{3}$$

where s is the specific charge of the (single) charge carrier species. We then introduce resistivity as a linear damping term, and an external electromotive force $\mathbf{K}$ as an addition to the electric field term:

$$\frac{\partial_t(\mathbf{J})}{s} = \rho_\mathrm{f}(\mathbf{E} + \mathbf{K}) + \mathbf{J} \times \mathbf{B} - \gamma \mathbf{J} \tag{4}$$

In an infinite wire of conductivity σ with uniform constant electric field $\mathbf{E}$ applied along it we should be able to find γ by recovering the classical proportionality $\mathbf{J}_\mathrm{f} = \sigma \mathbf{E}$ in the equilibrium state $\partial_t(\mathbf{J}_\mathrm{f}) = 0$. The magnetic field $\mathbf{B}$ is assumed to be zero.

$$\begin{aligned}0 &= \rho_\mathrm{f}\mathbf{E} - \gamma \mathbf{J}\\ \mathbf{J} &= \frac{\rho_\mathrm{f}\mathbf{E}}{\gamma} = \sigma \mathbf{E}\\ \gamma &= \frac{\rho_\mathrm{f}}{\sigma}\end{aligned} \tag{5}$$

Here ρ_f is not the dynamic scalar field but rather the constant free charge density of the conductor when there are no disturbances. To account for the movement of the charge carriers, we add a $-\frac{\mathbf{J}}{\rho_\mathrm{f}} \cdot \nabla \mathbf{J}$ term, similarly to a material derivative:

$$\partial_t(\mathbf{J}) = s(\rho_\mathrm{f}(\mathbf{E} + \mathbf{K}) + \mathbf{J} \times \mathbf{B} - \gamma \mathbf{J}) - \frac{\mathbf{J}}{\rho_\mathrm{f}} \cdot \nabla \mathbf{J} \tag{6}$$

We then make use of the continuity equation to constrain the time evolution of ρ_f:

$$\partial_t(\rho) = -\nabla \cdot \mathbf{J} \tag{7}$$

This concludes our treatment of the free charges inside the conductor.

III. Discretisation

We use s collocated grid for all vector fields for simplicity. So far the state of the system is $[\mathbf{B}, \mathbf{E}, \mathbf{J}, \rho]$, and the time evolution equations are

$$\begin{aligned}\partial_t(\mathbf{E}) &= c^2(\nabla \times \mathbf{B} - \mu_0 \mathbf{J})\\ \partial_t(\mathbf{B}) &= -\nabla \times \mathbf{E}\\ \partial_t(\mathbf{J}) &= s(\rho_\mathrm{f}(\mathbf{E} + \mathbf{K}) + \mathbf{J} \times \mathbf{B} - \frac{\rho_\mathrm{f}}{\sigma}\mathbf{J}) - \frac{\mathbf{J}}{\rho_\mathrm{f}} \cdot \nabla \mathbf{J}\\ \partial_t(\rho) &= -\nabla \cdot \mathbf{J},\end{aligned} \tag{8}$$

We also have to ensure that the divergence macroscopic Maxwell's equations are satisfied for the system at all times:

$$\begin{aligned}\nabla \cdot \mathbf{E} &= \rho\\ \nabla \cdot \mathbf{B} &= 0\end{aligned} \tag{9}$$

We might also need to worry about limiting ρ_f to a realistic range of values - so that there are locally not fewer than zero conduction electrons and so that the conduction band is not locally "overfull".

IV. Integration

The time evolution equations we arrived at previously are

$$\begin{aligned}\partial_t(\mathbf{E}) &= c^2(\nabla \times \mathbf{B} - \mu_0\mathbf{J}) \\ \partial_t(\mathbf{B}) &= -\nabla \times \mathbf{E} \\ \partial_t(\mathbf{J}) &= s(\rho_\mathrm{f}(\mathbf{E}+\mathbf{K}) + \mathbf{J}\times\mathbf{B} - \frac{\rho_\mathrm{f}}{\sigma}\mathbf{J}) - \frac{\mathbf{J}}{\rho_\mathrm{f}}\cdot\nabla\mathbf{J}\end{aligned} \tag{10}$$

Where the $\partial_t(\mathbf{J})$ equation applies only inside a conductor, s is the specific charge of the (single) charge carrier species of the conductor, ρ_f is the (assumed constant) free charge density of the conductor, $\mathbf{K}$ is the externally applied electromotive force (as a force per unit charge rather than a potential difference), and σ is the conductivity of the conductor. The $\partial_t(\rho_\mathrm{f})$ equation can be dropped for now, as it does not affect the rest.

Let the state of the system be $\mathbf{S}_\mathrm{n} = \mathbf{S}(\mathrm{n}\delta\mathrm{t}) = \begin{pmatrix}\mathbf{E}_\mathrm{n}\\ \mathbf{B}_\mathrm{n}\\ \mathbf{J}_\mathrm{n}\end{pmatrix}$, where n is the time index. To use the Crank-Nicolson Leap-Frog method, we separate $\frac{\partial\mathbf{S}}{\partial t}$ into the linear component

$$\mathbf{F} = \begin{pmatrix} c^2(\nabla\times\mathbf{B} - \mu_0\mathbf{J}) \\ -\nabla\times\mathbf{E} \\ s\rho_\mathrm{f}(\mathbf{E} - \frac{1}{\sigma}\mathbf{J})\end{pmatrix}, \tag{11}$$

and the non-linear (with K being external, independent of the state) component

$$\mathbf{G} = \begin{pmatrix} 0 \\ 0 \\ s(\rho_\mathrm{f}\mathbf{K} + \mathbf{J}\times\mathbf{B}) - \frac{\mathbf{J}}{\rho_\mathrm{f}}\cdot\nabla\mathbf{J}\end{pmatrix}. \tag{12}$$

Now using the Crank-Nicolson Leap-Frog method, we take the approximation

$$\frac{\mathbf{S}_{\mathrm{n}+1} - \mathbf{S}_{\mathrm{n}-1}}{2\Delta t} \approx \mathbf{G}_\mathrm{n} + \frac{\mathbf{F}_{\mathrm{n}+1} + \mathbf{F}_{\mathrm{n}-1}}{2}. \tag{13}$$

Rearranging,

$$\mathbf{S}_{\mathrm{n}+1} - \mathbf{F}_{\mathrm{n}+1}\Delta t = \mathbf{S}_{\mathrm{n}-1} + 2\mathbf{G}_\mathrm{n}\Delta t + \mathbf{F}_{\mathrm{n}-1}\Delta t \tag{14}$$

$$\begin{aligned}&\begin{pmatrix}\mathbf{E}_{\mathrm{n}+1} - c^2\Delta t(\nabla\times\mathbf{B}_{\mathrm{n}+1} - \mu_0\mathbf{J}_{\mathrm{n}+1}) \\ \mathbf{B}_{\mathrm{n}+1} + \Delta t\nabla\times\mathbf{E}_{\mathrm{n}+1} \\ \mathbf{J}_{\mathrm{n}+1} - s\rho_\mathrm{f}\Delta t(\mathbf{E}_{\mathrm{n}+1} - \frac{1}{\sigma}\mathbf{J}_{\mathrm{n}+1})\end{pmatrix} = \\ &= \mathbf{S}_{\mathrm{n}-1} + 2\mathbf{G}_\mathrm{n}\Delta t + \mathbf{F}_{\mathrm{n}-1}\Delta t\end{aligned} \tag{15}$$

V. Conclusion

References